DREW

Lecture 8: Digital Logic

CSCI 260 Computer Systems & Architecture
Fall 2024

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09/18/2024

One Level Up: Circuits

Combinational Logic Circuit:

- Output depends only on the *current input*.
- Do not have state/"memory".
- Used for *Decision Structures*.

Decoder
Multiplexer (MUX)
Full Adder

Let's talk about storage first.

R-S Latch
Gated D-Latch
Registers

Sequential Logic Circuit:

- Output depends on the *sequence of input* (past and present).
- Have states/"memory" that store (finite) info. from the past.
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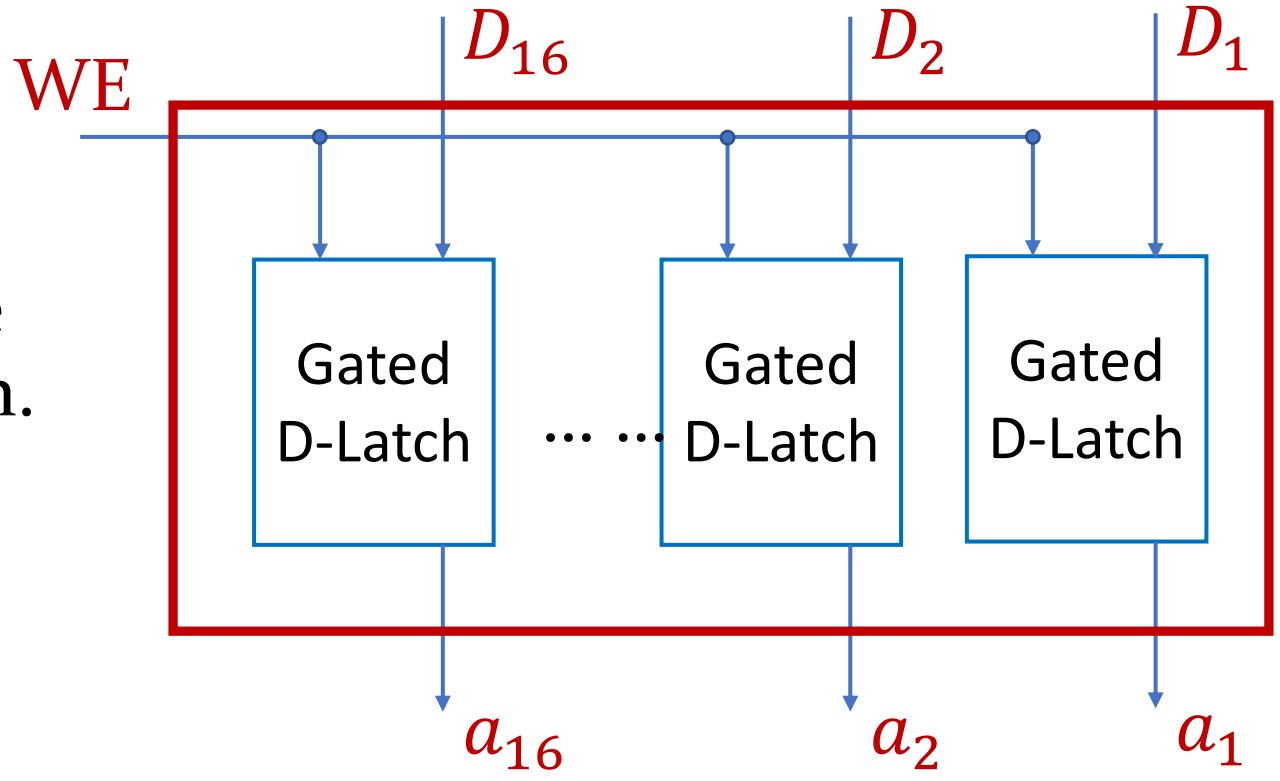
Register

- A *Register* can store an n-bit value, by combining n Gated D-Latches, where a single WE controls the write permission for all bits.

- The register width defines the *Word Length* in a processor.

- Is the smallest storage unit on the processor-level of abstraction.

- Used for temporarily storage of various internal info.

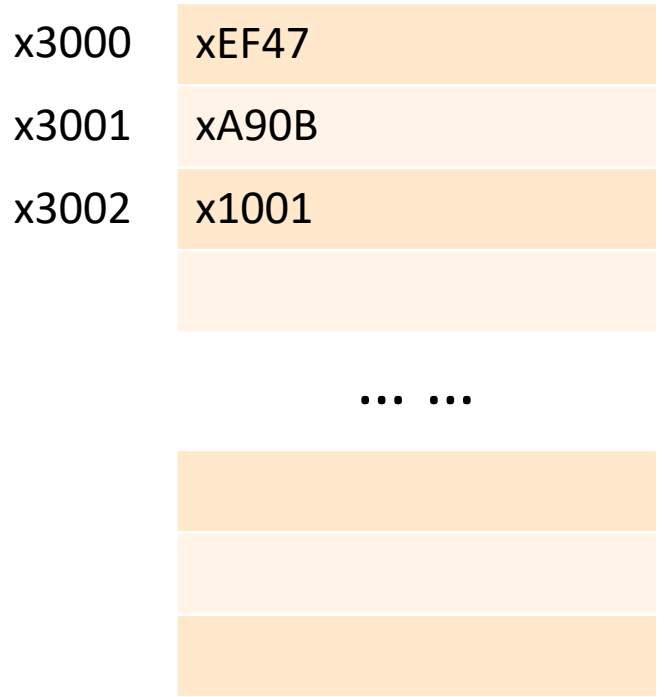


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§3.5 Memory

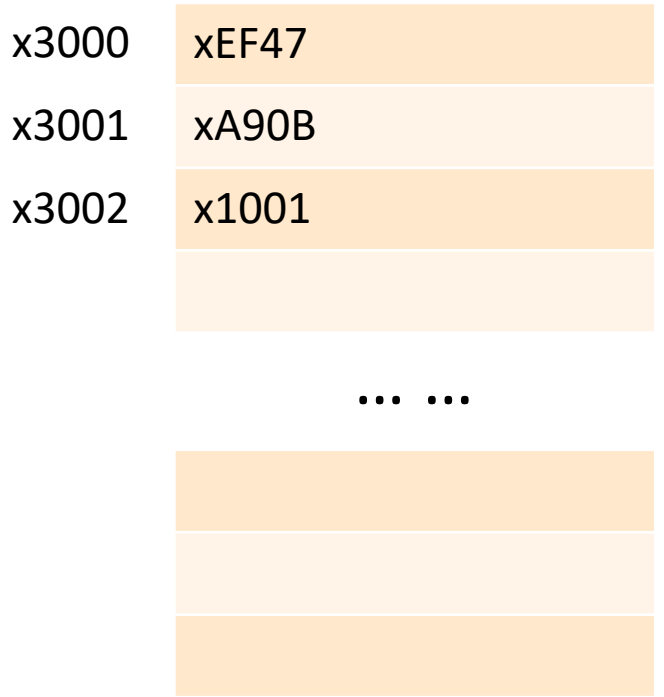
- From storing bit(s), move to storing multiple such sequences.
- **Idea**: Instead of keep going wide, let's extend the "depth".

- A computer memory is an $k \times m$ logical array:
  - m is the width, where each word has a unique id.: the *address it resides*;
  - thus, $n = 2^k$ is the *number of (unique) addresses*.
  - Each memory location, is then *m-addressable*.
    - Usually byte-addressable



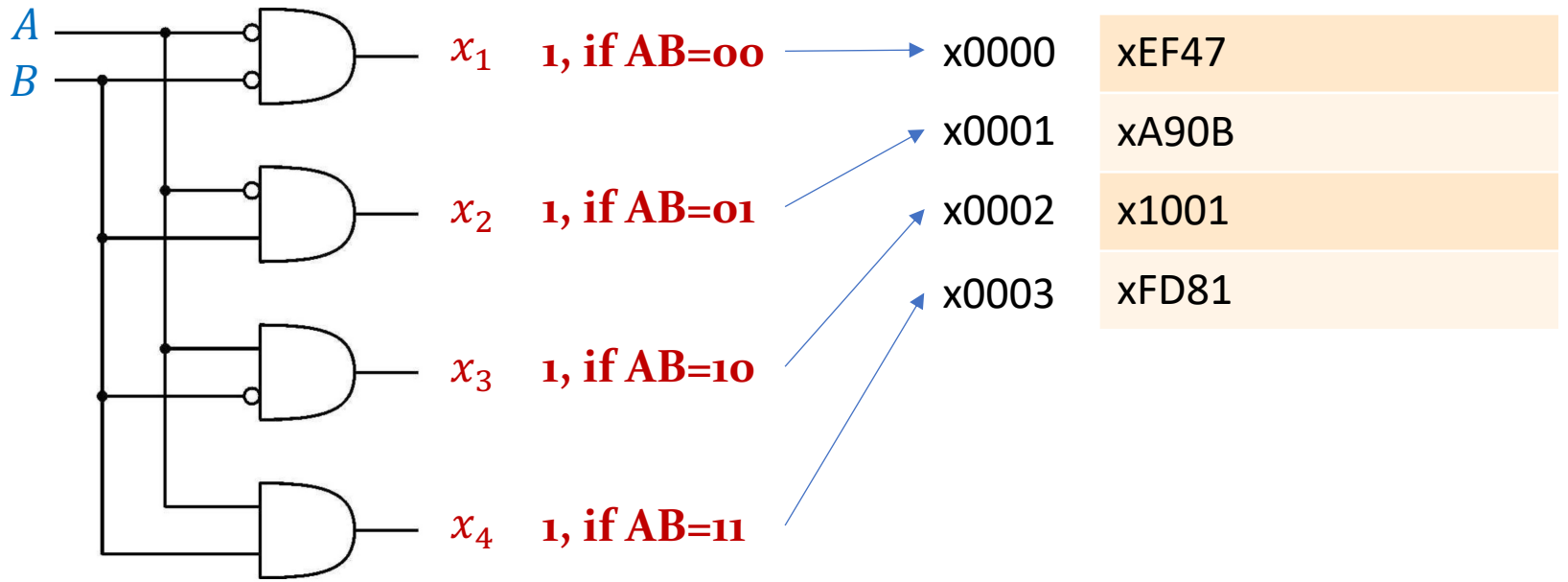
Memory

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  - $n = 2^k$ is the *number of (unique) addresses*.
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    - Usually byte-addressable
  - For a Byte-addressable (1 byte/address) computer, its memory size is equal to n byte. (2^{20} = 1GB, 2^{30} = 1TB, etc.)
  - Otherwise, also times the width in byte.



Memory

- To create a memory, we consider two dimensions
  - Within each address, 1 bit is a Gated D-Latch.
  - Each address need m such latches with one WE.
  - Need a k-bit ($n = 2^k$) input decoder to translate the locations.

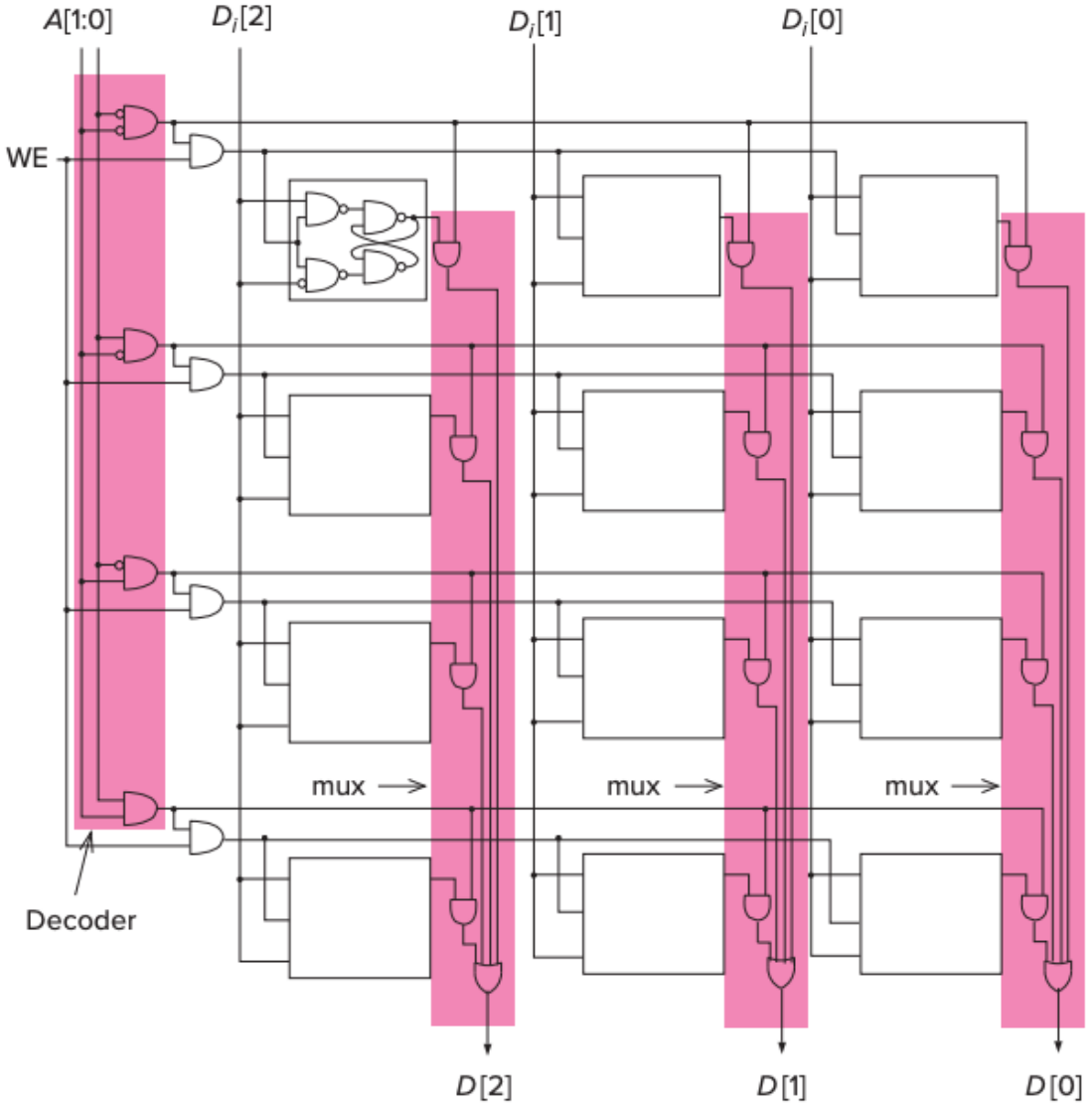


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Memory

- **Ex.**: a 2×3 memory

In reality, same structure is employed, but much more efficient/dense, with fewer transistors.



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Memory

Different types of *Main* memories

- Volatile: loses data after powered off.
  - *RAM* (Random Access Memory): fast, for storing freq. hit items (Caching).
    - Static RAM (SRAM): fastest, uses flip-flops!
    - Dynamic RAM (DRAM)
- Non-Volatile: not losing data after powered off.
  - ROM (Read-only Memory)

In the CPU
Register: fastest

Secondary
SSD (Solid-State Drive)
HDD (Hard Drive Disk)

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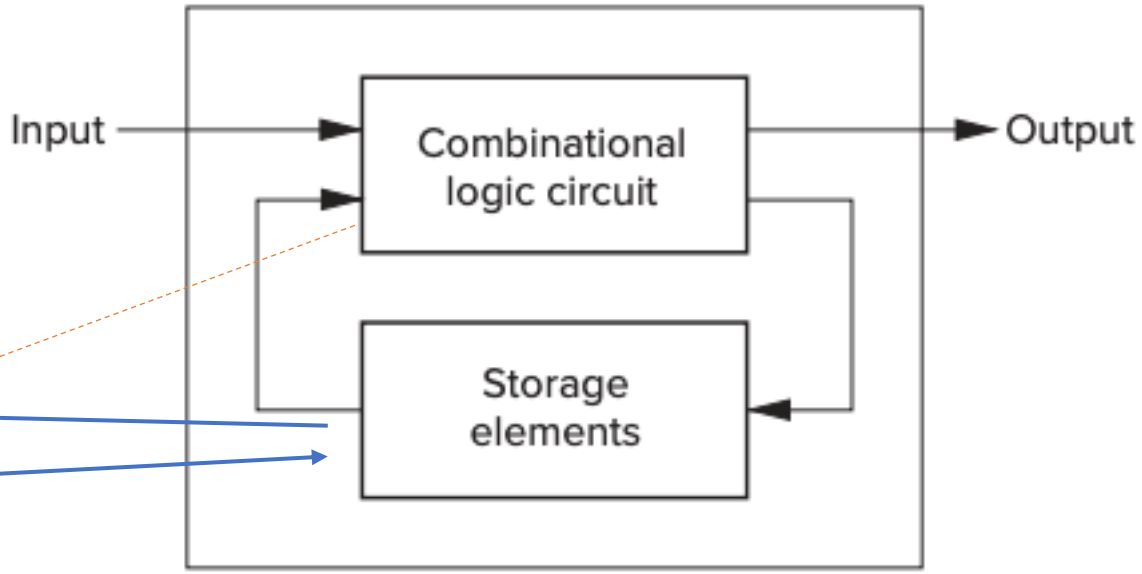
Sequential Logic Circuit:

- Output depends on the *sequence of input* (past and present).
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§3.6 Sequential Logical Circuits

- Structures processing info. based on the past that are stored in the circuits.

- Can use a *State* to represent a snapshot of the current system where all relevant info. are stored in that state.



- **Ex.**: Tic-Tac-Toe

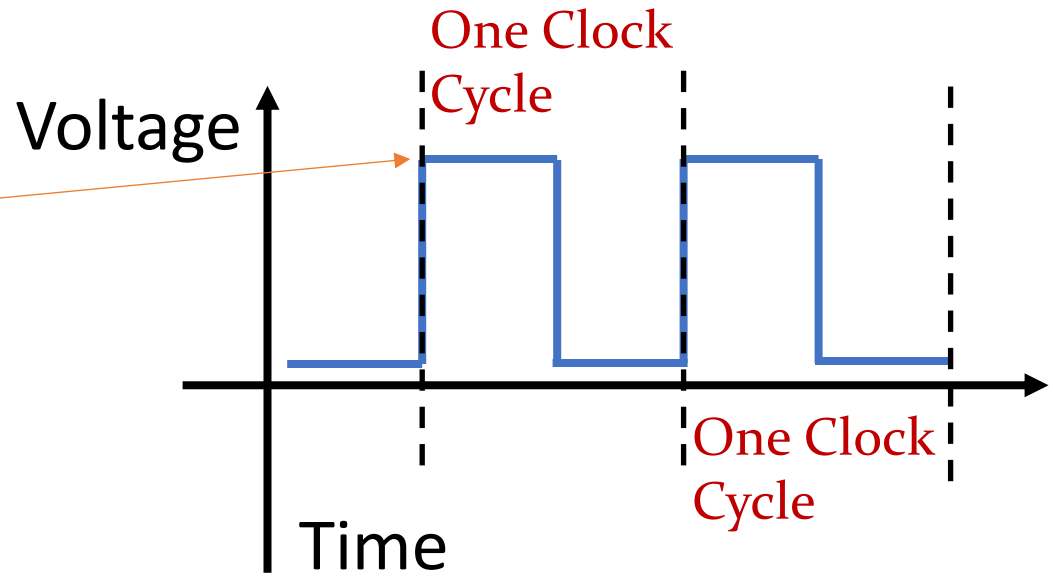
  State 0: Nothing on the board.
  State 1: Place O in the middle.
  State 2: Place X in the upper-left.
  …
  State k: Place O in the bottom-left.
  O wins, since the anti-diagonal has 3 Os.

Many more examples:
Sequential Lock,
Traffic Sign, Vending Machine,
etc.

Sequential Logical Circuits

- For computers, a *clock circuit* triggers the transition of states.

- A *Clock Cycle* is the time between two adjacent pulses, in which the computer moves from one state (at the peak) to another.

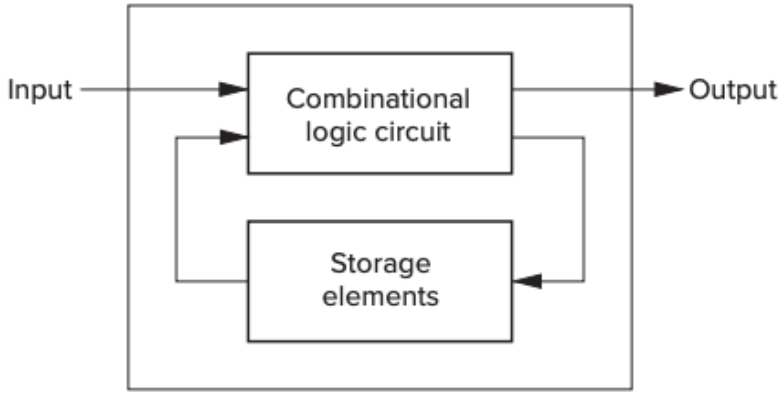


Clock cycles for *CPU Speed*

- 1 clock cycle per second is 1 Hz (Hertz).
- Chips nowadays 3.2 GHz is ~3.2×10^9 clock cycle/s.
- Will come back to what that means.

Sequential Logical Circuits

- Given this model of interaction, we can build *Finite State Machines* to solve for a certain set of problems.
  - Combinational: Determines action
  - Storage: Maintaining state



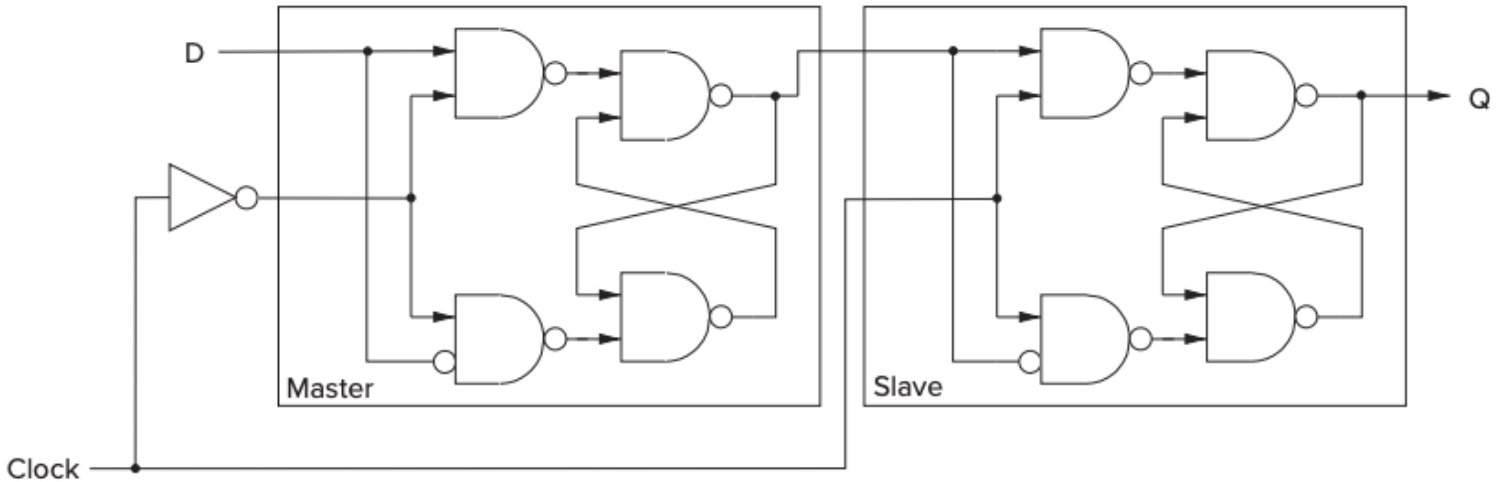
✓Combinational: implemented by logical gates

- One issue with storage:

  the change in a circuit is too quick, which interrupts the machine from reading the stored value before it is changed.

Mater/Slave Flip-Flop

- Solution: Master/Slave *Flip-Flop*: 2 connected Gated D-Latches with WE being the clock.

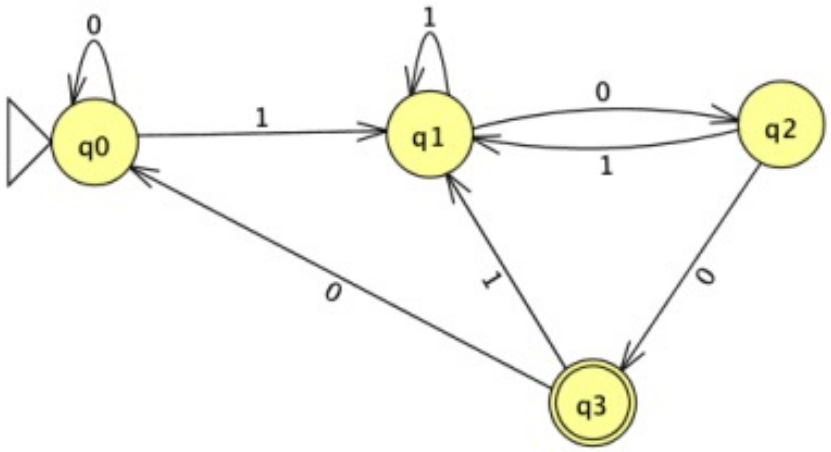


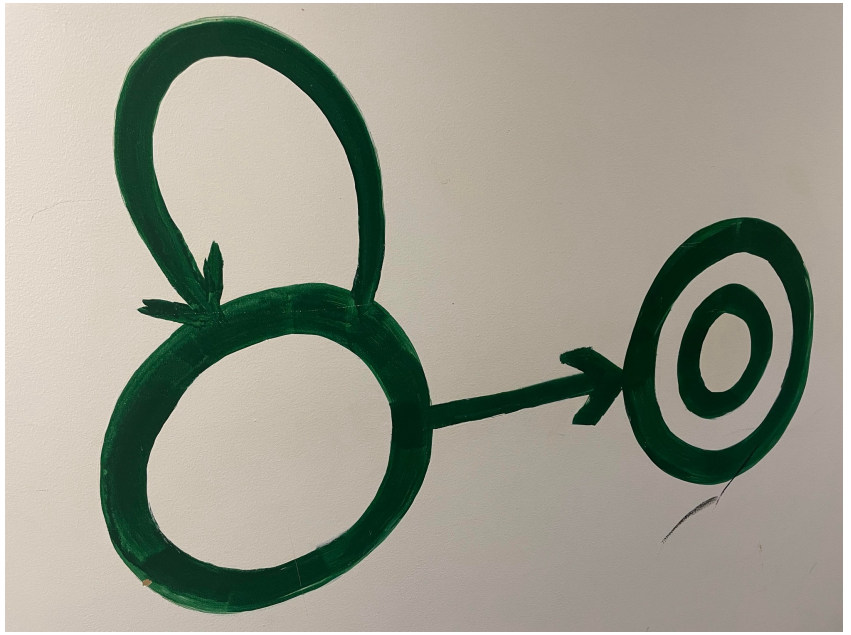
- The output of the master is the input to the slave.
- What's important is that, the WE at both latches are complement; i.e., they write at different time instances.
- Usually, more useful in our context.

Finite State Machine

- **Ex.**: (Topics course on Automata (FSM, PDA, TM) in the future)

FSM: Binary string when divided by 8
with remainder being 4.





Looks familiar?

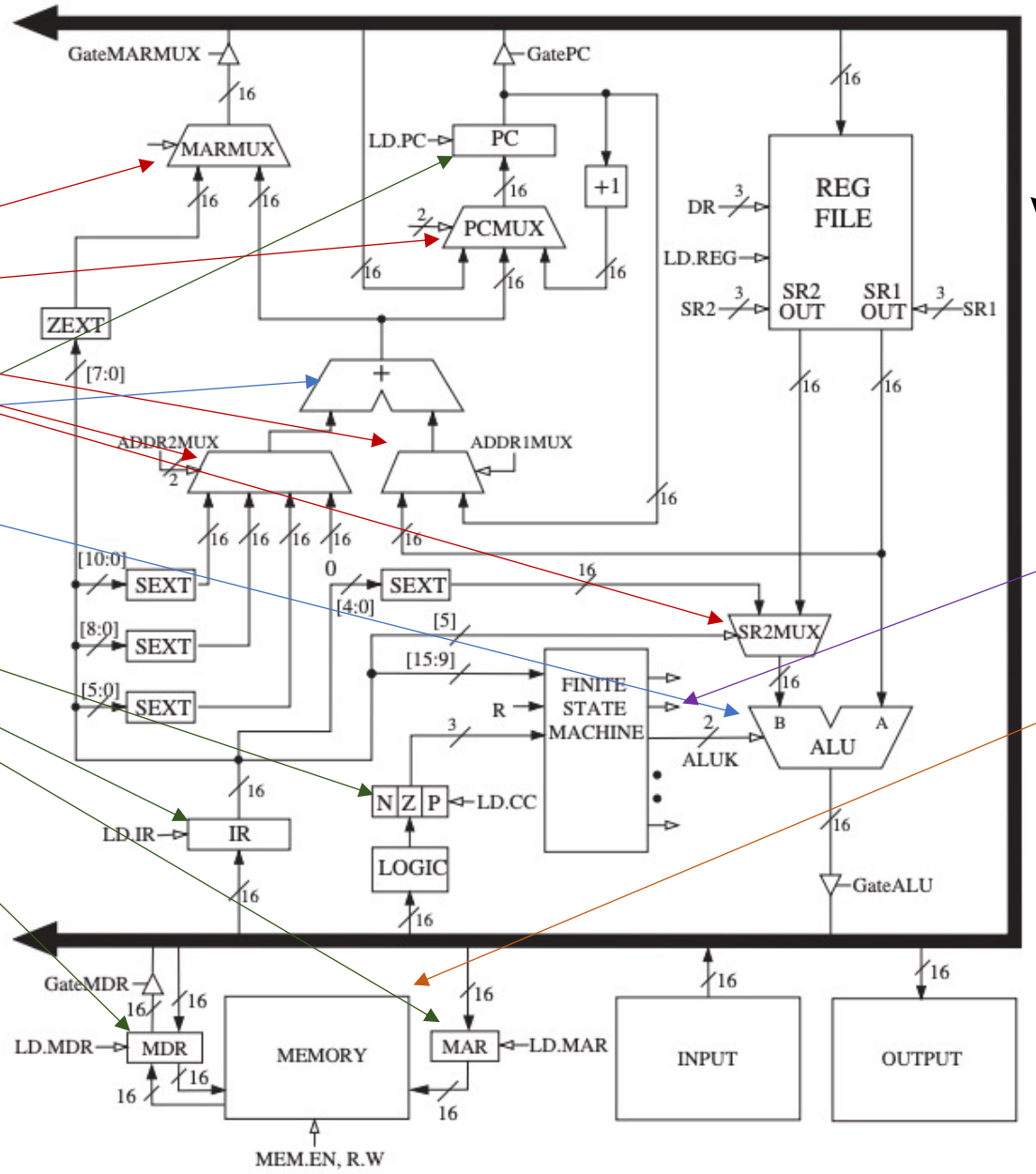
§3.7 Data Path of the LC-3: Where are we now?

- Little-Computer 3

MUX

Adder/ALU

Register



Bus

FSM

Memory

DREW

Before we end

- Here's something fun: https://everycircuit.com/circuit/6711990626287616/binary-to-7-segment-display-decoder

- This is converting a 4-bit binary sequence (in switches) to the corresponding hexadecimal representation (in 7-segment display).

Questions?

- **Next Week**:
  - The von Neumann Model
  - The LC-3 Assembly Language

- **HW 2** due on 09/27 (Fri.) by 11:59pm

- Now In-Class Exercise